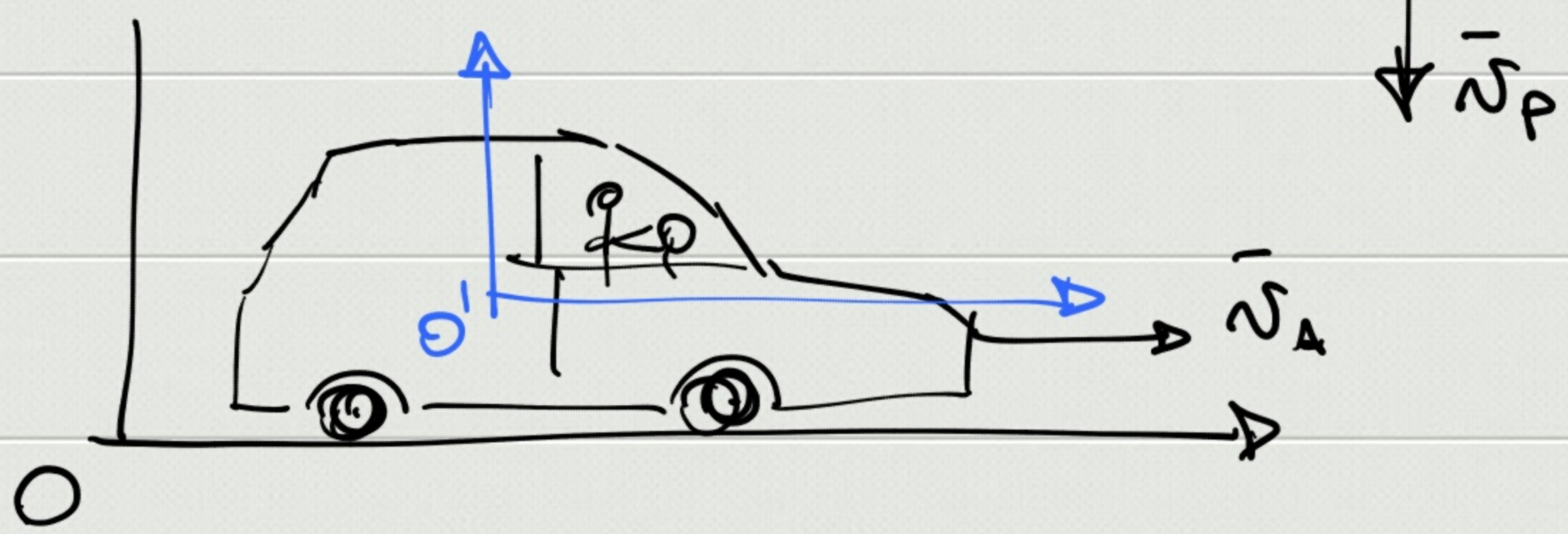
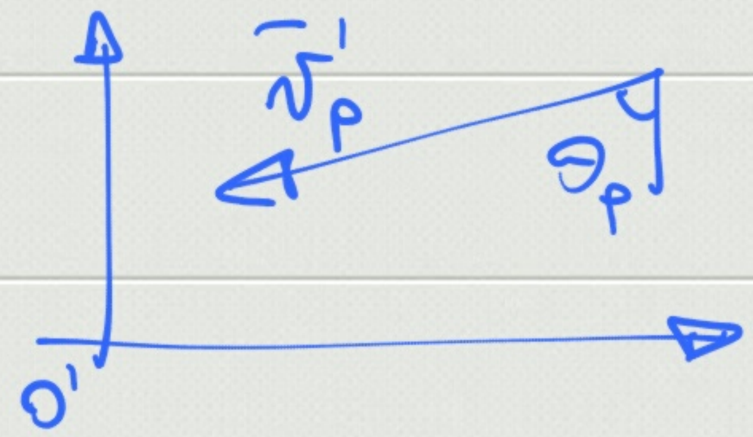
$v_A = 80\ km/h$

$\theta_p = 80°$

$|\bar{v}_p| = ?$



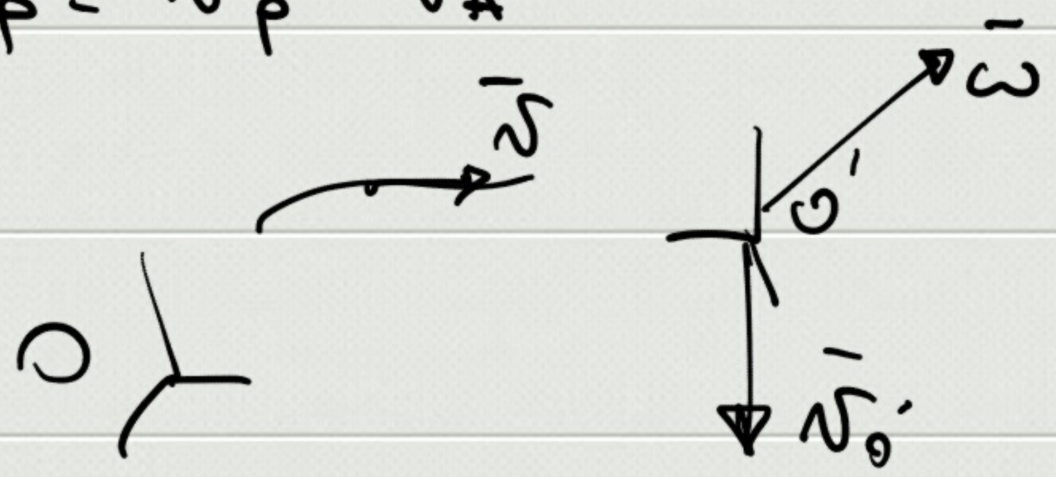


$v_p = v'_p + v_A$

$\bar{v}_p = \bar{v}'_p + \bar{v}_A \quad *$

$\bar{v}_p = \bar{v}'_p - \bar{v}_A$

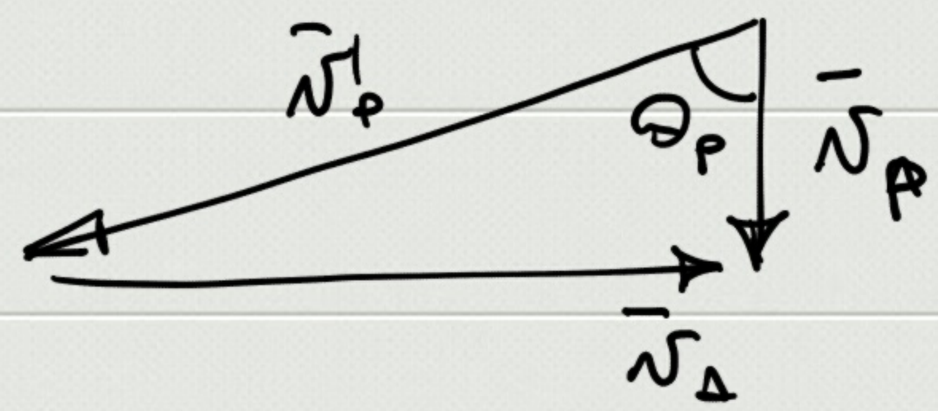
$v_p = v'_p - v_A$



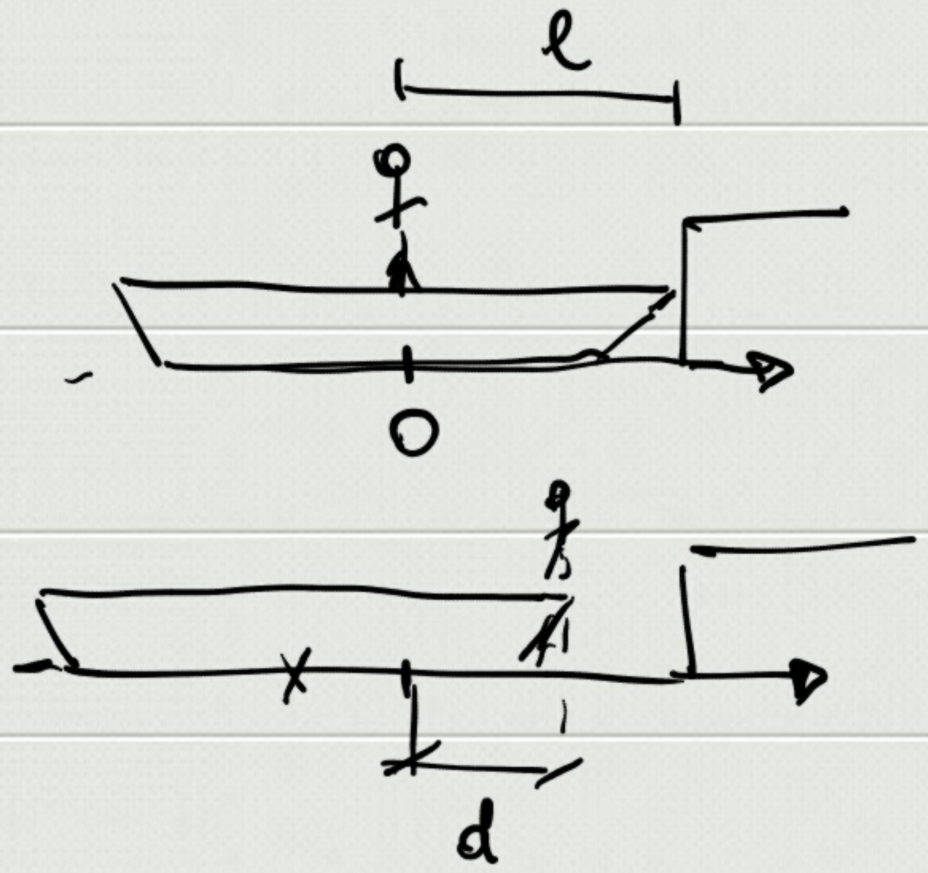
$$\boxed{\bar{v} = \bar{v}' + \bar{v}_{O'} + \bar{\omega} \times \bar{r}'}$$

$\bar{v}_p \quad \bar{v}'_p \quad \bar{v}_A \quad 0$

$\Rightarrow \bar{v}_p = \bar{v}'_p + \bar{v}_A$



$v_p = v_A \operatorname{ctg} \theta_p = 14.1\ km/h$



$m_c = 45\ kg \qquad v_{c,0} = 0$

$m_u = 90\ kg \qquad v_{u,0} = 0$

$l = 1.8\ m$

$d = ?$

$$\bar{R}^e = 0 \Rightarrow \bar{P} = m_u \bar{v}_u + m_c \bar{v}_c = cost = 0 \quad \Big\} \; v_{CM} = 0$$

$$\overset{!}{=} m_{TOT} \bar{v}_{CM}$$

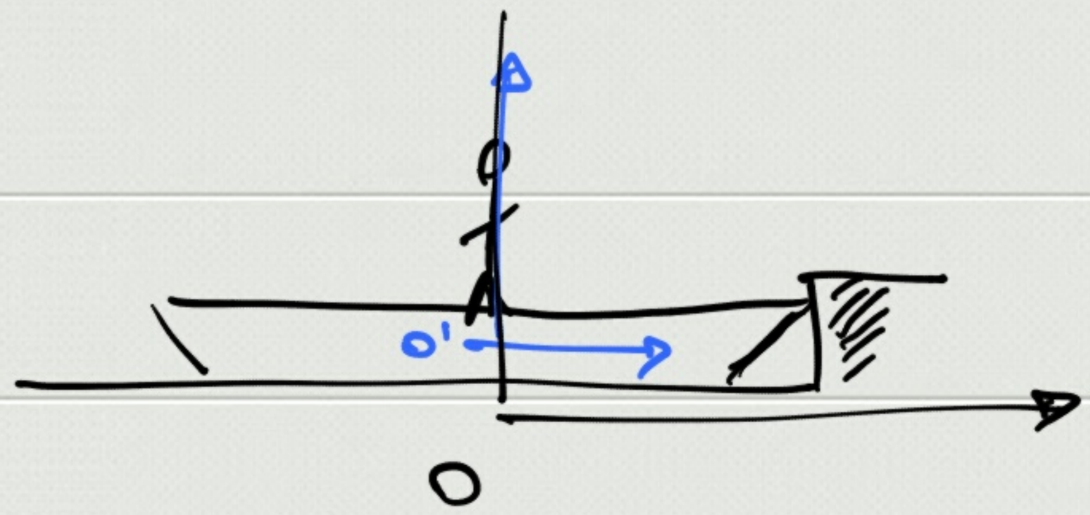
$$\Rightarrow x_{CM} = cost$$

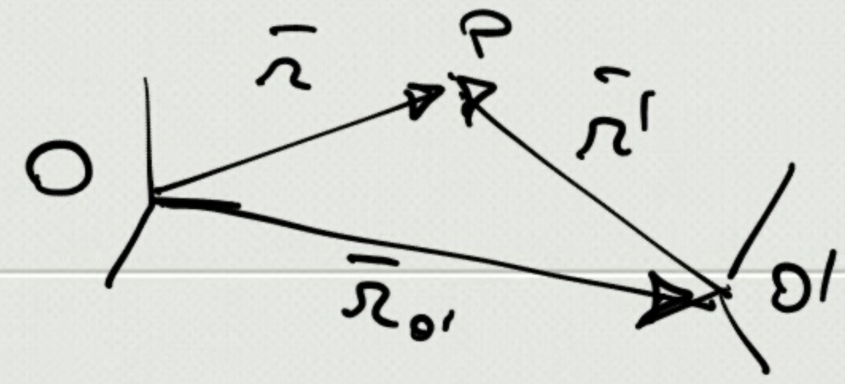
$$x^i_{CM} = \frac{m_u \cdot \emptyset + m_c \cdot \emptyset}{m_u + m_c} = 0$$

$$x^f_{CM} = \frac{m_u d + m_c (d - l)}{m_u + m_c}$$

$$\Rightarrow (m_u + m_c) d = m_c l$$

$$\Rightarrow d = \frac{m_c}{m_u + m_c} l = 0.6\ m$$



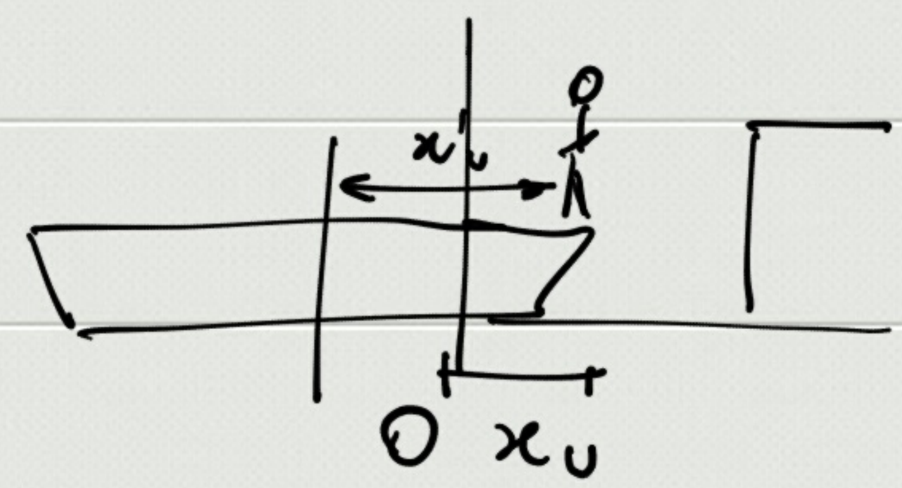


$$\boxed{\bar{r} = \bar{r}' + \bar{r}_{O'}}$$

$$\boxed{\bar{R}^{\varepsilon} = 0} \Rightarrow \boxed{\bar{P} = \text{cost} = 0} \Rightarrow x_{CM} = \text{cost} = 0$$

$$= m_U x_U + m_C x_C$$

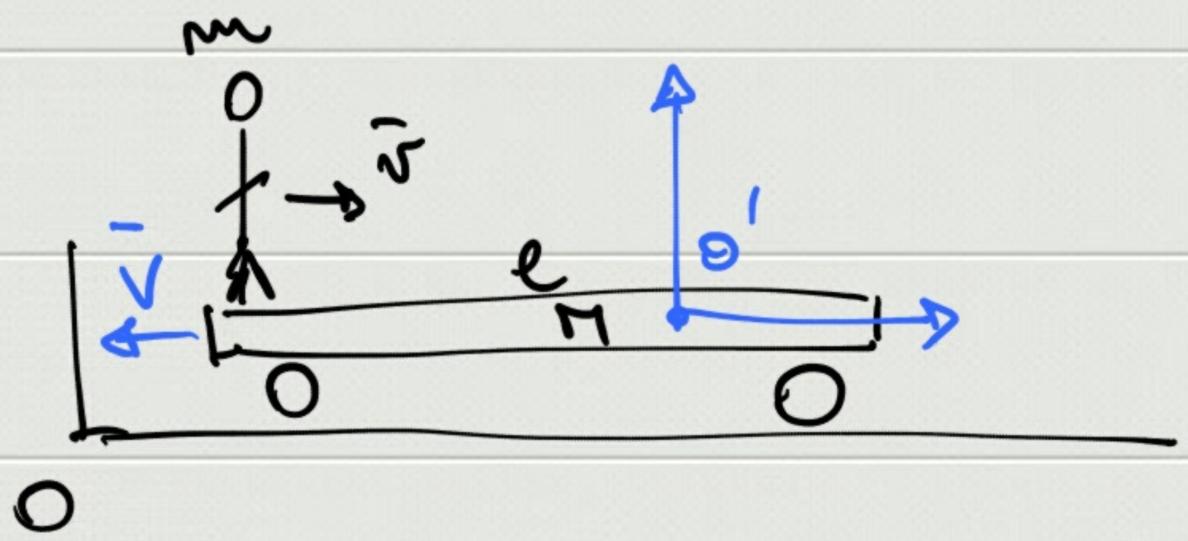
$$\begin{cases} m_U x_U + m_C x_C = 0 \\ x_U = \underbrace{x'_U}_{\ell} + x_C \end{cases}$$



$$\Rightarrow x_C = -\frac{m_U}{m_C} x_U$$

$$\Rightarrow x_U = \ell - \frac{m_U}{m_C} x_U \Rightarrow x_U\left(1 + \frac{m_U}{m_C}\right) = \ell$$

$$\Rightarrow x_U = \frac{m_C}{m_C + m_U} \ell$$



m , $M = 9m$

$\ell = 4\,m$

$v_0 = 0$

$a = 0.5\ m/s^2 \Rightarrow a'$

$V_f = ?$

$$\boxed{\bar{R}^E = 0} \Rightarrow \left.\begin{array}{l} \bar{P} = cost = 0 \\ \overset{!}{=} m\bar{v} + M\bar{V} \end{array}\right\} \Rightarrow \boxed{\bar{v} = -\frac{M}{m}\bar{V}}$$

$$\bar{v} = \bar{v}' + \bar{V}$$

SdR mobile : $v_f'^2 = \overset{0}{v_i'^2} + 2\overset{a}{a'}\overset{\ell}{\ell'} \Rightarrow v_f' = \sqrt{2a'\ell'}$

$$\begin{cases} v = -\frac{M}{m}V \Rightarrow v_f = -\frac{M}{m}V_f \\ v_f = \sqrt{2a'\ell'} + V_f \end{cases} \Rightarrow$$

$$-\frac{M}{m}V_f = \sqrt{2a'\ell'} + V_f \Rightarrow V_f = -\frac{\sqrt{2a'\ell'}}{1+\frac{M}{m}}$$

$$\Rightarrow V_f = -\frac{m}{m+M}\sqrt{2a'\ell'} = -0.2\ m/s$$